

Kitab al-Mukhtasar fi Hisab al-Jabr wa-l-Muqabala

li-Muhammad ibn Musa al-Khwarizmi

(taqriban 850–780 M)

al-Tab'a al-Arabiyya al-Muhaqqqa al-Mu'azzaza

Enhanced Arabic Scholarly Edition

al-Nass al-Asli ma'a Ta'liqat Tahqiqiyya wa-Shuruh Riyadiyya

Allifa hadha al-Kitab 'inda Balat al-Khalifa al-Ma'mun fi Baghdad

Hawali Sanat 200 H / 820 M

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'An Hadhihi al-Tab'a

Tuqaddimu hadhihi al-tab'a al-Kitab al-Asli fi 'ilm al-Jabr li-Muhammad ibn Musa al-Khwarizmi, alladhi allafahu fi Balat al-Khalifa al-Ma'mun fi Baghdad hawali sanat 820 M. Yashtamil hadha al-Isdar al-Mu'azzaz 'ala al-nass al-Arabi al-asli ma'a ta'liqat tahqiqiya mufassala wa-sharh riyadi wa-tawdihat hindisiya wa-siyag tarikhi – kulluha bi-l-lugha al-Arabiyya.

al-Nass al-asli ma'khudh min al-tab'a al-tahqiqiya li-'Ali Mustafa Musharrafa wa-Muhammad Mursi Ahmad (al-Qahira, 1359 H / 1939 M), allati istanadat ila al-makhtuta al-wahida al-mahfuza fi maktabat Bodleyan bi-Oxford (Hunt. ,214 nusikhat sanat 743 H / 1342 M).

Contents

Taqdimat al-Tab'a al-Mu'azzaza	5
Kalamat al-Musannif: al-Muqaddima al-Asliyya	7
1 fi al-A'dad wa-al-Anwa' al-Thalatha	9
1.1 al-Anwa' al-Thalatha min al-A'dad fi al-Jabr	9
2 al-HaSalat al-Sitt al-Kanuniyya lil-Mu'adlat	11
2.1 al-HaSalat al-Basita al-Thalath	11
2.1.1 al-HaSala al-Ula: al-mal yusawi usuLan	12
2.1.2 al-HaSala al-Thaniya: al-mal yusawi 'adadan	12
2.1.3 al-HaSala al-Thalitha: al-usuL tusawi 'adadan	13
2.2 al-HaSalat al-Murakkaba al-Thalath	14
2.2.1 al-HaSala al-Rabi'a: mal wa-usuL yusawi 'adadan	14
2.2.2 al-HaSala al-Khamisa: mal wa-'adad yusawi usuLan	15
2.2.3 al-HaSala al-Sadisa: usuL wa-'adad yusawi malan	16
3 al-Tashrihat al-Hindisiyya lil-HaSalat al-Murakkaba	17
3.1 Tashrih al-HaSala al-Rabi'a: "Mal wa-'asharatu usuL yusawi tis'a wa-thalathin" . . .	17
3.2 Tashrih al-HaSala al-Khamisa: "Mal wa-21 yusawi 10 usuL"	19
3.3 Tashrih al-HaSala al-Sadisa: 3" usuL wa-4 yusawi malan"	20
4 fi Darb al-Ta'abirat al-Jabriyya	21
4.1 Amthila 'ala al-Darb	22
5 fi al-Jam' wa-al-TarH	23
5.1 'Amaliyyat 'ala al-Juthur al-tarb'iyya	23
5.2 fi muDa'afat al-Juthur wa-niSfiha	23
6 fi al-Qisma	25
7 al-sittu Masa'il al-TawDiHiyya	27
al-mas'ala al-ula	27
al-mas'ala al-thaniya	28
al-mas'ala al-thalitha	28
al-mas'ala al-rabi'a	28
al-mas'ala al-khamisa	28
al-mas'ala al-sadisa	29
8 masa'il mukhtalifa	31
masa'il tawDiHiyya iDafiyya	31

I al-Tatbiqat al-'amaliyya	33
9 fi al-mu'amalat al-tijariyya	37
10 al-misaHa wa-l-handasa	39
10.1 al-muthallathat	39
10.2 al-dawa'ir	40
10.3 al-aHjam	40
10.4 al-muthallath al-qaaim al-zawiya (mubarhanat fifaghorus)	41
11 fi al-wasaya wa-'ilm al-fara'iD	43
fi ras al-mal wa-l-duyun	43
fi naw' akhar min al-wasaya	44
masa'il al-wasaya al-murakkaba	44
A mulakHHas al-HaSalat al-sitt	47
B mu'jam al-muSTalaHat al-jabriyya al-Arabiyya	49
mulahadha bibllyughrafiyya	51

Taqdimat al-Tab'a al-Mu'azzaza

Tu'tabar ummat al-Islam min aghna al-umam turathan 'ilmiyyan, wa-qad Tatawaqqalat 'alayha hadarat mukhtalifa min fajr al-tarikh ila al-yawm. Wa-fi kulli hadara minha qamat bi-qasT wafir min wajibiha al-'ilmi nahwa al-usra al-bashariyya. Wa-laysa yakfi an natahaddath 'an majdina al-'ilmi kama law kana usTura, bal yajib an yazhar hadha al-majd fi sura malmusa tara'ha al-a'yun wa-tunaluha al-aydi. Li-dhalika kana min al-muhimm an na'na bi-nashr al-kutub allati wada'aha aba'una wa-ajdaduna, khususan idha kanat hadhihi al-kutub hamma al-athar fi takyif al-tafkir al-bashari. Wa-la shakka anna fi muqaddimati hadhihi al-kutub **Kitab al-Khwarizmi fi al-Jabr wa-l-Muqabala**.

Ahimmiyyat Hadha al-Kitab fi Tarikh al-'Ilm

Kitab al-Khwarizmi fi al-Jabr wa-l-Muqabala huwa awwal kitab munaZZam fi 'ilm al-jabr 'ala al-itlaq. Wa-huwa yamthilu al-laHZa allati taHawwala fiha al-jabr min majmu'at min al-qawa'id al-munfasila ila 'ilm riyadi munazzam lahu usuluhi wa-qawa'iduhu al-khassa. Laqad adkhala al-Khwarizmi niZaman lil-mu'adlat min al-daraja al-thaniya bina'an 'ala thalathat anwa' min al-kamiyyat: **al-mal** (al-majhul al-tarb'i), wa-**al-jadhr** (al-majhul al-khatti), wa-**al-'adad** (al-thabit). Wa-min hadhihi al-anwa' al-thalatha is-tanbaTa **sitt haSalat** kanuniyya taghatti kulli ashkal al-mu'adlat al-tarb'iyya dhat al-ma'amilat al-mujiba.

Mustalahat al-Jabr al-Asasiyya:

- **al-Jabr** (al-jabr): idafat ma yanquSu min ahadi Tarfay al-mu'adla ila al-Taraf al-akhar li-istikmal al-tawazun
- **al-Muqabala** (al-muqabala): hadf ma yatashabahu min al-Tarfayn, ay Tarq al-mutashabihat min kila Tarfi al-mu'adla li-tashil
- **al-Mal** (al-mal): mirab' al-majhul, ay x^2
- **al-Jadhr** (al-jadhr): al-majhul dhatuhu, ay x
- **al-'Adad** (al-'adad): al-hadd al-thabit fi al-mu'adla

Kalamat al-Musannif: al-Muqaddima al-Asliyya

al-Nass al-Asli li-l-Khwarizmi

Bismi Allahi al-Rahmani al-Rahim.

Al-Hamdu lillahi 'ala ni'amihi li-ahli Ta'atihi, bi-ada'iha bi-qudratihi 'ala 'ibadihi al-mu'minina bihi, li-yuthayyibahum biha thawaban, wa-yuHafizahum biha 'an al-taghyir, wa-ya'rifahum qudratahu, wa-yudhillilahum li-'iZmatihi. Arsala Muhammadan Salla Allahu 'alayhi wa-Sallam bi-rislatihi ba'da inqiTa' al-rusul, wa-dhihab al-'adl, wa-fasad al-nuSul.

Wa-qad kana al-'ulama' fi al-azmani al-khaliya wa-fi al-umam al-salifa yu'allifuna al-kutub fi abwab al-'ulum wa-furu' al-ma'rifa, yatadhakkaruna min ba'dihim, wa-yarjuna al-ajra bi-qadri ma kana minhum, wa-yathiquna bi-ma yaHasil lahum min al-shukr wa-al-ijlal wa-al-'iZam, raDiina wa-law bi-qalilin min al-thina', qalilan muqarana bi-ma badhuluhu min al-jahd wa-al-mushaqqa fi kashf khafa'a al-'ilmi wa-izalat satayrihi.

Wa-dhalika al-Hubb li-l-'ilmi alladhi mayyaza Allahu bihi al-Imam al-Ma'mun Amir al-Mu'minin (ma'a ma manna bihi 'alayhi min al-khilafa bi-al-ijma', wa-al-libas alladhi albasahu iyahu, wa-al-zina allati zayyanahu biha), wa-dhalika al-laTf wa-al-tawaddud alladhi yubdihi li-l-'ulama', wa-tilka al-sur'a allati yuHamiha wa-yunSiha fi tawDiH al-mubhamat wa-izalat al-Su'ubat – qad Hamalni 'ala an u'allif mukhtaSan fi al-Jabr wa-l-Muqabala, aqtasiru fihi 'ala ma huwa aysar wa-anfa' fi al-Hisab, mimma yaHtaju ilayhi al-nasu da'iman fi masa'il al-mirath wa-al'aTaya wa-al-qisma wa-al-khuSumat wa-al-tijara wa-kulli ma yata'amaluna bihi fima bay-nahum, aw fi misaHat al-arDiyn wa-Hufri al-anhar wa-masa'il al-handasa wa-ghayriha min anwa' al-ashya' al-mukhtalifa al-aSnaf.

al-Ta'liq al-Tahqiqi:

Yastahillu al-Khwarizmi kitabahu bi-al-basmala al-taqlidiyya, thumma yuthni 'ala al-Khalifa al-Ma'mun (218–198) H / 833–813 M), wa-huwa min ashhura khulafa' Bani al-'Abbas tashji'an lil-'ilm. Laqad assasa al-Ma'mun **Bayt al-Hikma** fi Baghdad, wa-ja'alah markazan 'alamiyyan lil-baHth al-'ilmi, wa-jama'a fihi al-mutarjimina wa-al-'ulama' min al-muslimina wa-ghayrihim.

Yubayyinu al-Khwarizmi dafi'ahu li-ta'lif hadha al-Kitab: al-Haja al-mulahha li-uslub manZum fi Hal al-masa'il al-'amaliyya allati yuwajihuha al-nas yawmiyyan – min qismat al-tarakat ila misaHat al-aradi wa-Hufri al-qanawat. Wa-hadha yuwiZZahu anna al-jabr 'inda al-Khwarizmi lam yakun mujarrad niZariyya mujarrada, bal kana **adatan 'amaliyya** muWajahatan nahwa Hall al-mushkilat al-waqi'iyya.

al-Imam al-Ma'mun Amir al-Mu'minin (833–786) M):

Kana al-Ma'mun min akthar al-khulafa' tashji'an lil-'ilm wa-al-'ulama'. Amara bi-tarjamat al-kutub al-

Yunaniyya wa-al-Hindiyya wa-al-Suryaniyya ila al-Arabiyya, wa-assasa Bayt al-Hikma fi Baghdad allati kanat 'aSima al-'ilmi fi al-'alam. Wa-fi 'ahdihi taqallada al-Khwarizmi mansaBan fi Bayt al-Hikma, haythu allafa hadha al-kitab wa-ghayrahu min al-muSannafat fi al-riyadiyyat wa-al-falak wa-al-jughrafiya.

Chapter 1

fi al-A'dad wa-al-Anwa' al-Thalatha

al-Nass al-Asli li-l-Khwarizmi

Lamma naZartu fima yaHtaju ilayhi al-nasu fi Hisabatihim, wajadtu anna dhalika kulluhu 'adad. Wa-wajadtu aydan anna kullu 'adadin mu'allaf min al-waHadat, wa-annahu la yakhlu 'adad min an yanqasim ila waHadat. Wa-'alimtu anna kulla 'adadin yu'adda min al-waHid ila al-'ashara yazidu 'ala alladhi qablahu bi-waHid, thumma yuDa'afu al-'ashru wa-yuthallathu, kama fu'ila bi-l-waHadat, fa-yakunu 'ishrun wa-thalathun wa-haka-dha ila al-ma'iyya, thumma yuDa'afu al-ma'iyya wa-yuthallathuha kama fu'ila bi-l-'asharat wa-l-waHadat ila al-alf, thumma yukarru al-alfu kadhaka fi kulli 'adadin murakkab, wa-haka-dha ila aqSa Hadin fi al-'add.

al-NiZam al-'Adadi 'inda al-Khwarizmi:

Yabda'u al-Khwarizmi kitabahu bi-munaqasha falsafiyya li-l-'adad, mustakhdiman al-niZam al-'adadi al-'ashari alladhi yu'rafu al-yawma bi-"al-arqam al-Arabiyya" (wa-hiya fi al-aSl min aSl Hind). Yasifu al-Khwarizmi al-niZam al-marakaZi li-l-a'dad Haythu kullu mawDi' yamthilu quwwa min 'ashara. Hadhihi al-muqaddima laysat majarrad tamhid, bal hiya i'lan 'an **al-niZam al-'adadi al-mawDi'i** alladhi sa-yaj'alu kull al-Hisabati al-taliqa mumkina.

Laqad istaw'abba al-Khwarizmi anna kulla masa'ala Hisabiyya – sawa' fi al-tijara aw al-misaHa aw al-mirath – yumkinu irja'uha fi al-nihaya ila mafhum al-'adad, wa-min thumma yumkinu Halluha bi-l-Turuq al-riyadiyya.

1.1 al-Anwa' al-Thalatha min al-A'dad fi al-Jabr

al-Nass al-Asli li-l-Khwarizmi

Fa-wajadtu anna al-a'dada allati yaHtaju ilayha fi Hisab al-Jabr wa-l-Muqabala thalathatu anwa': **juthur**, wa-**amwal**, wa-**a'dad mufarrada** la tat'allaqu bi-jadhrin wa-la bi-malin.

Amma al-jadhr fa-huwa kullu miqdarin yuDribu fi nafsahi mu'allafan min waHadatin aw a'dadin sa'ida aw kusurin nazila. Wa-amma al-mal fa-huwa jami'u ma Hasala min Darb al-jadhr fi nafsahi. Wa-amma al-'adad al-mufrad fa-huwa kullu 'adadin yudhkaru bi-ghayri nisba ila jadhrin wa-la malin.

al-Anwa' al-Thalatha bi-l-Rumuz al-Haditha:

al-Isem al-Arabi	al-Ramz al-Hadith	al-Mathal	al-Ta'rif
al-mal) x^2 (	mal	x^2	mirab' al-majhul
al-jadhr) x (	jadhr	x	al-kamiyya al-majhula
al-'adad) c (	'adad	39	al-thabit al-'adadi

Mulahadha: kalimat “**mal**” fi al-iStilaH al-jabri 'inda al-Khwarizmi ta'ni *mirab' al-majhul*. Wa-amma “**al-shay**” (alladhi yumathtilu “al-jadhr”) fa-huwa al-majhul al-khatti. Wa-l-jadhr huwa alladhi tun-sha'u minhu al-amwal bi-l-Darb fi al-nafs. Wa-l-'adad al-mufrad huwa kullu 'adadin la yatagayyaru bi-taghyyuri al-majhul.

Tafsir al-Mustalahat al-Jabriyya:

Kalimat “**al-jadhr**” fi hadha al-siyag tuKhTallifu 'an al-jadhr al-tarb'i kama nafhamuhu al-yawm. Fa-l-jadhr 'inda al-Khwarizmi huwa **al-majhul dhatuhu**, ay x fi al-rumuz al-Haditha. Wa-l-mal huwa **tarb' hadha al-majhul**, ay x^2 .

Wa-qad ikhtara al-Khwarizmi kalimat “**mal**” rubbama li-annana mirab' al-majhul yumathtilu *al-tharwa* aw *al-mumtalakat* fi al-tafkir al-iqtiSadi li-l-'asr – fa-huwa al-kamiyya al-akbar wa-al-ahamm fi al-mu'adala.

Chapter 2

al-HaSalat al-Sitt al-Kanuniyya lil-Mu'adlat

al-Nass al-Asli li-l-Khwarizmi

Hadhi-hi hiya al-HaSalat al-sitt alladhi dhakartuha fi muqaddimati hadha al-kitab. Thalathun minha la taHtaju ila an tunaSafa al-juthuru, wa-qad bayyantu kayfa tujab. Wa-amma al-thalath al-ukhra allati yajibu fiha naSf al-juthuri, fa-ara an ubayyinaha aydan mufassaLan ma'a dhikri al-shakl li-kulli haSala yubayyin sabab al-naSf.

YuQassimu al-Khwarizmi jami'a al-mu'adlat al-tarb'iyya ila **sitt haSalat kanuniyya** – thalath *basita* wa-thalath *murakkaba*. Hadha al-taqsimu yughatti jami'a ashkal al-mu'adlat al-tarb'iyya dhat al-ma'amilat al-mujiba. Wa-l-sabab fi al-iqtiSir 'ala al-ma'amilat al-mujiba huwa anna al-Khwarizmi yafhamu kulla Haddin fi al-mu'adala 'ala annahu **kamiyya hindisiyya** (Tul aw misaHa), wa-la yumkinu an takuna al-misaHa saliba.

al-Tamayyuz bayn al-HaSalat al-Basita wa-l-Murakkaba:

- al-HaSalat al-basita (al-ula wa-l-thaniya wa-l-thalitha): la taHtaju ila **naSf al-juthur** – yumkinu Halluha bi-l-tashil al-mubashshar.
- al-HaSalat al-murakkaba (al-rabi'a wa-l-khamisa wa-l-sadisa): tataTallaBu **naSf al-juthur**, wa-hadhihi hiya al-HaSalat allati yaZharu fiha **takmil al-murabba'** (completing the square).

2.1 al-HaSalat al-Basita al-Thalath

al-HaSalat al-Basita (la tatallaBu naSf al-juthur):

al-HaSala	al-Mu'adala	al-Hall	al-Isem al-Arabi
al-Ula	$ax^2 = bx$	$x = b/a$	mal yusawi usuLan
al-Thaniya	$ax^2 = c$	$x = \sqrt{c/a}$	mal yusawi 'adadan
al-Thalitha	$bx = c$	$x = c/b$	usuL tusawi 'adadan

2.1.1 al-HaSala al-Ula: al-mal yusawi usuLan

al-Nass al-Asli li-l-Khwarizmi

Mathal: mal yusawi khamsata usuLihi. Jadhr al-mal khamsa, wa-l-mal khamsa wa-'ishrun, wa-huwa yusawi khamsata a'da' jadhr.

Mathal akhar: thulthu mal yusawi arba'ata usuLin. Fa-l-mal kulluh yusawi ithnay 'ashara jadhra; ay mi'a wa-arba'a wa-arba'in, wa-jadhrhu ithna 'ashar.

Mathal thalith: khamsatu amwalin tusawi 'ashara usuLan. Fa-mal wahid yusawi jadhrayn; jadhr al-mal ithnan, wa-malahu arba'a.

al-Tawdih al-Riyadi lil-HaSala al-Ula:

$$x^2 = 5x \Rightarrow x = 5, \quad x^2 = 25$$

$$\frac{1}{3}x^2 = 4x \Rightarrow x^2 = 12x = 144, \quad x = 12$$

$$5x^2 = 10x \Rightarrow x^2 = 2x = 4, \quad x = 2$$

al-Tariqa al-'Amma: fi al-mu'adala $ax^2 = bx$, naqsimu al-Tarfayn 'ala ax (bi-afTariD $x \neq 0$) fa-nuHallu 'ala $x = b/a$. Amma Hall $x = 0$ fa-kana yutajahalu taqlidiyan li-annahu la ma'na 'amaliyya lahu fi muZamm al-masa'il.

2.1.2 al-HaSala al-Thaniya: al-mal yusawi 'adadan

al-Nass al-Asli li-l-Khwarizmi

Mathal: mal yusawi tis'a. Fa-hadha mal, wa-jadhrhu thalatha.

Mathal akhar: khamsatu amwalin tusawi thamanin. Fa-mal wahid yusawi khumus thamanin, wa-huwa sitta 'ashar.

Mathal thalith: niSf mal yusawi thamaniya 'ashar. Fa-l-mal sitta wa-thalathun, wa-jadhrhu sitta.

al-Tawdih al-Riyadi lil-HaSala al-Thaniya:

$$x^2 = 9 \Rightarrow x = 3$$

$$5x^2 = 80 \Rightarrow x^2 = 16, \quad x = 4$$

$$\frac{1}{2}x^2 = 18 \Rightarrow x^2 = 36, \quad x = 6$$

al-Tariqa al-'Amma: fi al-mu'adala $ax^2 = c$, naqsimu 'ala a fa-nuHallu 'ala $x^2 = c/a$, thumma na'khudhu al-jadhr al-tarb'i li-kila al-Tarfayn: $x = \sqrt{c/a}$.

2.1.3 al-HaSala al-Thalitha: al-usuL tusawi 'adadan

al-Nass al-Asli li-l-Khwarizmi

Mathal: jadhr wahid yusawi thalathata 'adadan. Fa-l-jadhr thalatha, wa-maluha tis'a.

Mathal akhar: arba'atu usuLin tusawi 'ishrina. Fa-jadhr wahid yusawi khamisa, wa-l-mal alladhi yunsha'u minhu khamisa wa-'ishrun.

Mathal thalith: niSf jadhr yusawi 'ashara. Fa-l-jadhr kulluh yusawi 'ishrina, wa-l-mal alladhi yunsha'u minhu arba'umi'a.

al-Tawdih al-Riyadi lil-HaSala al-Thalitha:

$$x = 3$$

$$4x = 20 \Rightarrow x = 5, \quad x^2 = 25$$

$$\frac{1}{2}x = 10 \Rightarrow x = 20, \quad x^2 = 400$$

al-Tariqa al-'Amma: fi al-mu'adala $bx = c$, naqsimu 'ala b fa-nuHallu 'ala $x = c/b$. Wa-hadhihi abSar al-HaSalat, li-annaha mu'adala khattiyya.

2.2 al-HaSalat al-Murakkaba al-Thalath

al-HaSalat al-Murakkaba (tatallaBu naSf al-juthur – takmil al-murabba'):

al-HaSala	al-Mu'adala	al-Isem al-Arabi
al-Rabi'a	$ax^2 + bx = c$	mal wa-usuL yusawi 'adadan
al-Khamisa	$ax^2 + c = bx$	mal wa-'adad yusawi usuLan
al-Sadisa	$bx + c = ax^2$	usuL wa-'adad yusawi malan

limadha tatallaBu hadhihi al-HaSalat naSf al-juthur?

li-annaha taShtamilu 'ala thalathat aTraf mal + juthur + 'adad, wa-la yumkinu 'azlu al-majhul bi-l-qisma al-mubashshara. badalan min dhalika, nastakhdimu tariqat **takmil al-murabba'** allati taHawwilu al-mu'adala al-tarb'iyya ila mu'adala min al-shakl $(x+p)^2 = q$ yumkinu Halluha bi-akhdhi al-jadhr al-tarb'i.

2.2.1 al-HaSala al-Rabi'a: mal wa-usuL yusawi 'adadan

al-Nass al-Asli li-l-Khwarizmi

Mathal: mal wa-'asharatu usuLihi tusawi tis'a wa-thalathin darhaman.

al-Hall: naSifu 'adada al-juthuri, fa-yakunu khamisa, wa-naDribuhu fi nafsih fa-yakunu khamisa wa-'ishrina, fa-nuDifuhu ila tis'a wa-thalathin fa-yakunu tis'a wa-sittin, fa-na'khudh jadhrhu fa-yakunu thamaniya, fa-nanquSu minhu naSf 'adadi al-juthuri wa-huwa khamisa, Yabqa thalatha, wa-hadha huwa jadhr al-mal al-maTlub, wa-l-mal tis'a.

al-Hall al-'Am lil-HaSala al-Rabi'a:

$$x^2 + bx = c \implies x = \sqrt{\left(\frac{b}{2}\right)^2 + c} - \frac{b}{2}$$

al-Tatbiq 'ala al-mathal: $x^2 + 10x = 39 \implies x = \sqrt{25 + 39} - 5 = \sqrt{64} - 5 = 8 - 5 = 3$

al-Tafsir: al-Khwarizmi la yaktubu Sighatan 'amma – bal yuqaddimu **Khwarizmiyya** (silsilat khuTwat) yumkinu tatbiQuha 'ala ay mu'adala min hadha al-shakl. Wa-hadha huwa al-sabab fi anna naSam-miyanfusana al-yawma "Khwarizmiyyat" (algorithms)!

limadha ya'malu hadha al-Hall?

Tariqat al-Khwarizmi ta'tamidu 'ala fikrat **takmil al-murabba'**. 'indama nuDifu $(b/2)^2$ ila kila Tarfi al-mu'adala $x^2 + bx = c$, naHsalu 'ala: $x^2 + bx + (b/2)^2 = c + (b/2)^2$

al-Taraf al-aysar huwa al-an mirab' kamil: $(x + b/2)^2 = c + (b/2)^2$

'inda akhdhi al-jadhr al-tarb'i: $x + b/2 = \sqrt{c + (b/2)^2}$

Wa-bi-Tarqi $b/2$: $x = \sqrt{(b/2)^2 + c} - b/2$

Wa-hadha bi-ZabT ma yaf'aluhi al-Khwarizmi, wa-in duna istikhdam al-rumuz al-jabriyya al-Haditha!

2.2.2 al-HaSala al-Khamisa: mal wa-'adad yusawi usuLan

al-Nass al-Asli li-l-Khwarizmi

Mathal: mal wa-wahid wa-'ishruna 'adadan yusawi 'asharata usuLihi.

al-Hall: naSifu 'adada al-juthuri fa-yakunu khamisa, wa-naDribuhu fi nafsiki fa-yakunu khamisa wa-'ishrina, fa-nanquSu minhu al-wahida wa-l-'ishrina al-mulHaqqayni bi-l-mal, Yabqa arba'a, fa-nastakhrij jadhrha fa-yakunu ithnayn, fa-nanquSuhu min naSf 'adadi al-juthuri wa-huwa khamisa, Yabqa thalatha, wa-hadha jadhr al-mal al-maTlub, wa-l-mal tis'a. Aw tuDifu al-jadhr ila naSf al-juthuri, fa-yakunu sab'a, wa-hadha jadhr al-mal al-maTlub, wa-l-mal tis'a wa-arba'un.

al-Hall al-'Am lil-HaSala al-Khamisa:

$$x^2 + c = bx \implies x = \frac{b}{2} \pm \sqrt{\left(\frac{b}{2}\right)^2 - c}$$

al-Tatbiq: $x^2 + 21 = 10x \Rightarrow x = 5 \pm \sqrt{25 - 21} = 5 \pm 2 = 3$ aw 7

mulahadha muhimma: hadhihi hiya al-HaSala **al-wahida** allati yakunu fiha Hallan – wa-li-dhalika fa-inna al-Khwarizmi yadhkuru SariHan annahu yumkinu istikhdam al-jam' aw al-TarH. Amma fi al-HaSalat al-ukhra fa-yakunu al-Hall wahidan faqaT.

shuruT imkaniyyat al-Hall fi al-HaSala al-Khamisa:

ba'da naSf al-juthur wa-Darb al-naSf fi nafsiki, idha kana al-natij **aqall** min al-'adad al-mulHaqq bi-l-mal, fa-l-masa'la **mumtani'a** (la Hall Haqiqiyyan). Amma idha kana al-natij **musawiyan** lil-'adad, fa-jadhr al-mal yusawi naSf al-juthur wafahu, bi-la ziyada wa-la nuqsan.

bi-l-rumuz al-Haditha: al-dakhil taHt al-jadhr yajib an la yakun saliban: $(b/2)^2 - c \geq 0$. Wa-hadha ya'ni $b^2 \geq 4c$, wa-huwa al-shart al-ma'ruf al-yawma bi-ism **al-mumayyiz** (discriminant) lil-mu'adala al-tarb'iyya.

2.2.3 al-HaSala al-Sadisa: usuL wa-'adad yusawi malan

al-Nass al-Asli li-l-Khwarizmi

Mathal: thalathatu usuLin wa-arba'atu a'dadin yusawi malan.

al-Hall: naSifu al-juthuri fa-yakunu wahidan wa-niSf, wa-naDribuhu fi nafsiki fa-yakunu ithnayn wa-rub'an, fa-nuDifuhu ila al-arba'a fa-yakunu sitta wa-rub'an, fa-nastakhrij jadhraka fa-yakunu ithnayn wa-niSfan, fa-nuDifuhu ila naSf al-juthuri wa-huwa wahid wa-niS fa-yakunu arba'a, wa-hadha jadhr al-mal, wa-l-mal sitta 'ashar.

al-Hall al-'Am lil-HaSala al-Sadisa:

$$bx + c = x^2 \implies x = \frac{b}{2} + \sqrt{\left(\frac{b}{2}\right)^2 + c}$$

al-Tatbiq: $3x + 4 = x^2 \Rightarrow x = 1\frac{1}{2} + \sqrt{2\frac{1}{4} + 4} = 1\frac{1}{2} + \sqrt{6\frac{1}{4}} = 1\frac{1}{2} + 2\frac{1}{2} = 4$

mulahadha: fi hadhihi al-HaSala yakunu al-Hall **mujaban da'iman**, li-annana najma'u al-jadhr al-tarb'i ma'a niSf al-ma'amil. Hadha yatawaafaqu ma'a Tabi'at al-masa'il al-'amaliyya allati kana yata'amalu ma'aha al-Khwarizmi.

Chapter 3

al-Tashrihat al-Hindisiyya lil-HaSalat al-Murakkaba

al-Nass al-Asli li-l-Khwarizmi

Hadhi-hi hiya al-HaSalat al-sitt alladhi dhakartuha fi muqaddimati hadha al-kitab. Qad wuD-DoHat al-an. Qad bayyantu anna thalathan minha la taHtaju ila naSf al-juthuri, wa-'allamtu kayfa yajibu Halluha. Wa-amma al-thalath al-ukhra allati fiha naSf al-juthuri Daruriyyun, fa-ara min al-muSlaH an uuwiHHahu bi-fuSulin munfasila, yu'ta fi kulli minha shakl yubayyin sabab al-naSf.

al-manhaj al-Hindisi lil-Khwarizmi:

Min ahamma ma yumayyizu kitab al-Khwarizmi annahu la yaktaSi bi-taqdim **Khwarizmiyya** (al-ijra' al-Hisabi) li-kulli HaSala, bal yuqaddimu aydan **al-tashrih al-Hindisi** (geometric demonstration) alladhi yuZhir *limadha* ta'malu hadhihi al-tariqa. hadha al-manhaj ya'ksi al-ta'thir al-Yunani – khususan Iqlidus – 'ala al-riyadiyyat al-Islamiyya.

al-Khwarizmi yastakhdimu **murabba'at wa-ajniha (mustatTilat)** li-tamthil al-kamiyyat al-jabriyya. al-mal huwa murabba', wa-l-juthur hiya mustatTilat mulHaqa bi-jawanib al-murabba', wa-l-a'dad hiya murabba'at ma'rufa al-misaHa. hadha al-tafsir al-Hindisi huwa alladhi yubirru qawa'idat "naSf al-juthur" – fa-hiya al-tariqa allati nuHawwilu biha al-shakl ghayr al-munTaZZim ila murabba' kamil.

3.1 Tashrih al-HaSala al-Rabi'a: "Mal wa-'asharatu usul yusawi tis'a wa-thalathin"

al-Nass al-Asli li-l-Khwarizmi

al-shakl al-mubayyin li-hadha huwa **murabba'**, aDla'uhu majhula. Yumanthilu al-mal, jadhr alladhi turidu ma'rifatahu. Hadha huwa al-shakl A B, wa-kullu Dila' min aDla'ih i yumkinu i'tibaru ahada juthurihi; wa-idha Darabta ahada hadhihi al-aDla' fi ayyi 'adadin, fa-inna miqdar dhalika al-'adadi yumkinu an yunZar ilayhi ka-'adadi al-juthuri allati tuDafu ila al-mal. kullu Dila' min aDla' al-murabba' yumanthilu jadhr al-mal; wa-kama fi al-misal kanat al-juthuru murtabiTa bi-l-mal, yumkinuna an na'khudha rub'a al-'ashara, ay thnan wa-niSf, wa-nulHiqaha bi-kulli min al-aDla' al-arba'a lil-shakl. Fa-ma'a al-murabba' al-asli A B, tulHaqu arba'at ajniha mutawaziyya al-aDla', kullu minha lahu Dila' min aDla' al-murabba' ka-Tul,

wa-'adadi thnayni wa-niSf ka-'arD; wa-hiya al-ajniha J, Z, T, K.

Ladayna al-an murabba' mutasawi al-aDla' majhul, walakin fi kulli min zawaya-hi al-arba' yanquSu laHZa murabba'a min thnayni wa-niSf maDruba fi thnayni wa-niSf. Li-nu'awiDa 'an hadha al-nuqsan wa-nukammila al-murabba', yajib an nuDifa (ila ma ladayna) arba' marat murabba'a thnayni wa-niSf, ay khamisa wa-'ishrun.

Wa-nahnu na'lamu (min al-mas'ala) anna al-shakl al-awwal, ayyuha al-murabba' al-mumaththil lil-mal, ma'a al-ajniha al-arba'a al-muHiTa bihi, al-mumaththila lil-'asharati juthur, yusawi tis'a wa-thalathina 'adadan. Fa-idha aSafnaha ila khamisa wa-'ishrina, allati hiya musawiyya lil-murabba'at al-arba' 'inda zawaya al-shakl A B, allati tukammilu al-murabba' al-kabir D H, fa-n'alamu anna hadha al-majmui' ya'kul sitta wa-arba'in. ahadu aDla' hadha al-murabba' al-kabir huwa jadhrhu, ay thamaniya. Fa-idha naqasna Di'fa rub'i al-'ashara, ay khamisa, min thamaniya, kama min Tarafayyi al-Dil' al-murabba' al-kabir D H, fa-yabqa min hadha al-Dil' thalatha, wa-hadha huwa jadhr al-mal, aw Dil' al-shakl al-asli A B.

al-Burhan al-Hindisi lil-HaSala al-Rabi'a:

al-shakl	=	al-misaHa
Murabba' A B	=	x^2
+ 4 mutawaziyat al-aDla'	=	$10x$
al-majmui' al-juz'	=	39
+ 4 murabba'at zawiya)2.5 × 2.5(	=	25
al-majmui' (Murabba' D H al-kabir)	=	64
Dila' al-Murabba' al-kabir	=	$\sqrt{64} = 8$
$x = 8 - 2 \times 2.5 = 8 - 5 =$	3	

al-tafsir al-mar'i: nabda'u bi-murabba' janibuhu x (yumanthil al-mal). NulHiq bi-kulli janib mustatTila Tuluhu x wa-'arDuha 2.5 (rub' al-'adad .(10 al-arba' mustatTilat tumanthil $10x$. al-misaHa al-kulliyya hatta al-an hiya $x^2 + 10x = 39$.

Li-nukammila al-murabba' al-kabir, naHtaju li-izafat arba' murabba'at Sagira 'inda al-zawaya, kullu minha $2.5 \times 2.5 = 6.25$. al-majmui' $4 \times 6.25 = 25$. al-an al-murabba' al-kabir lahu misaHa $39 + 25 = 64$, fa-Dil'uhu huwa 8. NunquS minhu 5 ('arDu mustatTilatayn) fa-nuHallu 'ala $x = 3$.

limadha "rub' " thumma "Di'fa rub' ?"

Yulahidhu al-qari'u anna al-Khwarizmi ya'khudhu **rub'a** 'adad al-juthuri (rub' al-'ishrin = (2.5 li-insha' al-ajniha al-arba'a, thumma yuTraHu **Di'fa** hadha al-rub'i min Dil' al-murabba' al-kabir. limadha? li-anna kulli janib min jawaniB al-murabba' yaHaSSulu 'alayhi *janiHan* (wahid min kulli janib). Fa-idha kana 'arDu kulli janiH huwa $b/4$, fa-inna al-majmui' al-muTraHu min kila Tarafayi al-Dil' huwa $2 \times (b/4) = b/2$, wa-huwa bi-ZabT "naSf 'adadi al-juthur" al-madhkur fi al-Khwarizmiyya!

Wa-haka-dha yuZhiru al-burhan al-Hindisi *sabab* naSf al-juthur – wa-huwa taHawwilu al-shakl ila murabba' kamil.

3.2 Tashrih al-HaSala al-Khamisa: “Mal wa-21 yusawi 10 usuL”

al-Nass al-Asli li-l-Khwarizmi

Numaththilu al-mal bi-murabba' A D, Tul Dil'ihī majhul ladayna. Ila hadha nulHiq mustatTila, 'arDuhu musawin li-ahadi aDla' al-murabba' A D, kayna al-Dil' H N. Hadha al-mustatTil huwa H B. Tul al-shaklayni ma'an yusawi al-KhaTT H J. Na'lamu anna Tuluha 'asharatu a'dadin; fa-kullu murabba'in lahu aDla' mutasawiya wa-zawaya mutasawiya, wa-ahad aDla'ihī maDruban fi wahida huwa jadhr al-murabba', aw maDruban fi ithnatayn fa-huwa Di'fu jadhr al-shakl al-batt.

Fakama wada'at al-mas'ala anna malan wa-wahidan wa-'ishrina 'adadan yusawi 'asharata juthuran, yumkinuna an nastanTiZu anna Tul al-KhaTT H J yusawi 'asharata a'dadin, idha al-KhaTT J D yumanthilu jadhr al-mal. Naqsimu al-an al-KhaTT J H ila niSfayn mutasawiyayn 'inda al-nuqTa W: fa-l-KhaTT W J idhun yusawi H W.

Wa-na'lamu anna al-ruba'i H B yumanthilu al-wahida wa-l-'ishrina 'adadan allati zidat 'ala al-murabba'. Wa-wujida anna al-murabba' M T kulluhu yusawi khamisa wa-'ishrina. Fa-idha naqasna min hadha al-murabba' M T al-ruba'iyayn H T wa-M R, alladhayn yusawiyān wahan wa-'ishrina, fa-yabqa murabba' Sagir K R, yumanthilu al-furq bayna khamisa wa-'ishrina wa-wahid wa-'ishrina. Wa-huwa arba'a; wa-jadhrh, al-mumaththal bi-l-KhaTT W A alladhi yusawi A J, huwa ithnan.

al-Burhan al-Hindisi lil-HaSala al-Khamisa:

$$\begin{array}{rcl}
 MT = (10/2)^2 & = & 25 \\
 HT + MR = HB & = & 21 \\
 KR = 25 - 21 & = & 4 \\
 \sqrt{KR} & = & 2 \\
 x = WJ - WA = 5 - 2 = & \boxed{3} & \\
 \text{aw } x = WJ + WA = 5 + 2 = & \boxed{7} &
 \end{array}$$

al-tafsir: Nabni murabba'an janibuhu x (al-mal). NulHiq bihi mustatTilan Tuluha 10 wa-'arDuhu x – walakin al-misaHa al-za'ida 21 tumaththalu bi-l-mustatTil alladhi yulSaqu kharīj al-murabba'. Nukamilu al-murabba' al-kabir bi-Tul 10 (jami' al-juthur), wa-na'khudhu niSfahu 5. al-murabba' al-kabir $5 \times 5 = 25$ (naqisan al-mustatTil)21(yu'tina murabba'an Sagiran)4(janibuhu 2. al-Hallani ya'tiyani min imkaniyyati TarHi aw jam'i hadha al-janib min al-niSf.

3.3 Tashrih al-HaSala al-Sadisa: 3“ usuL wa-4 yusawi malan”

al-Nass al-Asli li-l-Khwarizmi

Li-yakun al-mal mumaththan bi-ruba'i aDla'uhu majhula ladayna, wa-in kanat mutasawiya fima baynaha, kama al-zawaya. Hadha huwa al-murabba' A D, alladhi yashtamilu 'ala al-thalathati juthurin wa-al-arba'ati a'dadin al-madhkura fi hadhihi al-mas'ala.

Fi kulli murabba'in ahad aDla'ih, maDruban fi wahida, huwa jadhr. NaqTu' al-an min al-murabba' A D al-ruba'i H D, wa-na'khudh ahad aDla'ih H J li-thalatha, wa-huwa 'adad al-juthur. Wa-huwa musawin li-A R. Yatba'u idhan anna al-ruba'i H B yumanthilu al-arba'a a'dadan allati zidat 'ala al-juthur. al-an nunaSSifu al-Dil' J H alladhi yusawi thalathata juthur, 'inda al-nuqTa W; min hadha al-qisma nabni al-murabba' W T alladhi huwa hasilu Darb niSf al-juthur (wahid wa-niSf) fi nafsih, ay thnan wa-rub'.

al-Burhan al-Hindisi lil-HaSala al-Sadisa:

$$\begin{aligned}
 \text{Murabba' } WT &= (3/2)^2 &= 2.25 \\
 AN + KL &= AR = 4 &(\text{al-a'dad al-muDafa}) \\
 GM &= WT + (AN + KL) = 2.25 + 4 &= 6.25 \\
 AG &= \sqrt{GM} = \sqrt{6.25} &= 2.5 \\
 x &= AG + GW = 2.5 + 1.5 = &\boxed{4}
 \end{aligned}$$

al-tafsir: Nabda'u bi-murabba' kabir $x \times x = x^2$ (yumanthilu al-mal. NaqTa'u minhu mustatTila) $3 \times x = 3x$ (yumanthilu al-juthur. ma yabqa huwa al-mustatTil alladhi yumanthilu al-'adad .4 Nukammilu al-murabba' al-Sagir 'inda al-zawiya bi-niSf al-juthur .(1.5) Na'khudhu al-jadhr al-tarb'i lil-majmui' wa-nujmi' 'alayhi niSf al-juthur.

Chapter 4

fi Darb al-Ta'abirat al-Jabriyya

al-Nass al-Asli li-l-Khwarizmi

Sa-u'allimuka al-an kayfa taDribu al-a'dada al-majhula – ay al-juthur – ba'Duha fi ba'D, in kanat munfarida, aw idha uDifat ilayha a'dad, aw idha nuqiSat minha a'dad, aw idha nuqiyat min a'dad; aydan kayfa tujmi'uha ba'Duha ila ba'D, aw kayfa tanquSu ba'Duha min ba'D. Kullama urida Darbu 'adadin fi 'adadin, fayajib an yukarrara ahaduhuma bi-'adadi ma yaHtaw-ihi al-akhar min al-waHadat.

Idha kanat a'dadun akbaru mu'allafa ma'a waHadatin li-tuDafa aw tunqas minhum, fa-inna **arba'a Darbatin** lazima; ay: al-a'dad al-akbaru fi al-a'dad al-akbar, wa-l-a'dad al-akbar fi al-waHadat, wa-l-waHadat fi al-a'dad al-akbar, wa-l-waHadat fi al-waHadat.

Idha kanat al-waHadat al-mu'allafa ma'a al-a'dadi al-akbar mujibatayni kulltayhima, fa-l-Darbatu al-rabi'a mujiba; wa-idha kanat salibatayni kulltayhima, fa-l-Darbatu al-rabi'a mujiba aydan. Wa-idha kanat ihdahuma mujiba wa-l-ukhra saliba, fa-l-Darbatu al-rabi'a saliba.

Qawa'id al-Darb al-Arba'i 'inda al-Khwarizmi:

Yuqaddimu al-Khwarizmi huna qawa'id Darb dhati al-Haddayn – ay ma nusammi al-yawma “tawzi' al-Darb 'ala al-jam'.” Idda aradna Darba $(a + x)(b + y)$, fa-nahnu naHtaju li-arba' Darbat munfasila:

- .1 al-akbar fi al-akbar: $a \times b$
- .2 al-akbar fi al-waHadat: $a \times y$
- .3 al-waHadat fi al-akbar: $x \times b$
- .4 al-waHadat fi al-waHadat: $x \times y$

Wa-hadhihi hiya bi-ZabT al-Sigha allati na'rifuha al-yawma: $(a + x)(b + y) = ab + ay + bx + xy$

Ahammiyyat qawa'id al-Isharat: Yulahidu al-Khwarizmi anna “al-salibi fi al-salibi = mujib” wa-“al-salibi fi al-mujibi = salib”. Wa-hadhihi awwalu Siyaghat ma'rufa li-qawa'id al-isharat fi tarikh al-riyadiyyat!

Qawa'id al-Isharat fi al-Darb:

	mujib	salib
mujib	mujib	salib
salib	salib	mujib

$$(+x)(+y) = +xy$$

$$(-x)(-y) = +xy$$

$$(+x)(-y) = -xy$$

$$(-x)(+y) = -xy$$

hadhihi al-qawa'id kanat thawriyya fi zamaniha – fa-qad samahat bi-l-ta'amul ma'a "al-kamiyyat al-naqiSa" (al-a'dad al-saliba) fi siyaq al-Darb wa-l-qisma.

4.1 Amthila 'ala al-Darb

al-Nass al-Asli li-l-Khwarizmi

al-mithal al-awwal: 'ashara wa-waHid yuDribani fi 'ashara wa-ithnayn.

'asharat 'asharat mi'a; waHid 'ashara 'asharatu 'asharat mujiba; ithnan 'ashara 'ishruna mujiba; wa-waHid fi ithnayn ithnan mujiban; al-majmui' mi'a wa-ithnan wa-thalathun.

al-mithal al-thani: 'ashara naqis wahid, yuDrabu fi 'ashara naqis wahid.

Fa-'asharat 'asharat mi'a; al-salib wahid fi 'ashara 'asharatu 'asharat saliba; al-salib al-akhar fi 'ashara 'asharatu 'asharat salibat ukhra, fa-yakunu thamanun; lakinna al-salib wahid fi al-salib wahid wahid mujib, fa-yakunu al-natij wahidan wa-thamanin.

al-mithal al-thalith: 'ashara wa-shay' yuDribani fi 'ashara wa-shay'.

Fa-'asharat 'asharat mi'a, wa-'asharat shay' 'asharat ashya'; wa 'asharat shay' 'asharat ashya'; wa-l-shay' maDruban fi al-shay' mal mujib, fa-yakunu al-natij mi'at dirham wa-'ishruna shay'an wa-malan mujiban wahidan.

al-mithal al-rabi': 'ashara naqis shay', fi 'ashara naqis shay'.

Fa-'asharat 'asharat mi'a; wa-salib shay' fi 'ashara 'asharatu ashya' saliba; wa-salib shay' akhar fi 'ashara 'asharatu ashya' saliba. Lakin salib shay' maDruban fi salib shay' mal mujib. Fa-l-natij mi'a wa-mal, naqis 'ishruna shay'an.

al-mithal al-khamis: 'ashara naqis shay', fi 'ashara wa- ziyadat shay'.

Fa-taql: 'asharat 'asharat mi'a; wa-salib shay' fi 'ashara 'asharat ashya' saliba; wa-shay' fi 'ashara 'asharat ashya' mujiba; wa-salib shay' fi shay' mal mujib; fa-l-natij mi'at dirham, naqis malin.

al-Tawdih al-Riyadi li-l-amthila:

$$(10 + 1)(10 + 2) = 100 + 10 + 20 + 2 = 132$$

$$(10 - 1)(10 - 1) = 100 - 10 - 10 + 1 = 81$$

$$(10 + x)(10 + x) = 100 + 10x + 10x + x^2 = 100 + 20x + x^2$$

$$(10 - x)(10 - x) = 100 - 10x - 10x + x^2 = 100 - 20x + x^2$$

$$(10 - x)(10 + x) = 100 - 10x + 10x - x^2 = 100 - x^2$$

mulahadha: al-mithal al-khamis yuZhiru natijatan mudhishatan: $(10 - x)(10 + x) = 100 - x^2$ – wa-huwa farq al-mirabba'ayn (difference of squares), ihdha ahamm al-Sughar al-jabriyya fi jami' al-riyadiyyat!

Chapter 5

fi al-Jam' wa-al-TarH

5.1 'Amaliyyat 'ala al-Juthur al-tarb'iyya

al-Nass al-Asli li-l-Khwarizmi

i'lam anna jadhr mi'atayni naqis 'ashara, mudafan ila 'ishrina naqis jadhr mi'atayni, huwa 'asharat.

Wa-jadhr mi'atayni naqis 'ashara, munqasan min 'ishrin naqis jadhr mi'atayni, huwa thalathun naqis Di'fi jadhr mi'atayni; wa-Di'fu jadhr mi'atayni yusawi jadhr thamanimi'a.

al-Tawdih al-Riyadi:

$$(\sqrt{200} - 10) + (20 - \sqrt{200}) = 10$$

$$(20 - \sqrt{200}) - (\sqrt{200} - 10) = 30 - 2\sqrt{200} = 30 - \sqrt{800}$$

al-mulahadha al-muhimma: al-Khwarizmi ya'rifu anna $2\sqrt{200} = \sqrt{800}$. Wa-hadha yatba'u min al-qawa'id al-'amma $a\sqrt{b} = \sqrt{a^2b}$. Fa-l-jadhr al-tarb'i yu'ammuqu kakyan jabri mustaqill yumkinu jam'uhu wa-TarQuhu mitla al-mutaghayyirat.

5.2 fi muDa'afat al-Juthur wa-niSfiha

al-Nass al-Asli li-l-Khwarizmi

Idha aradta an tuDa'ifa jadhr ayi malin ma'lum aw majhul, (bi-ma'na annaka taDribuhu fi ithnayn) fa-yakfi annaka taDriba ithnayn fi ithnayn, thumma fi al-mal; fa-jadhr al-natij yusawi Di'fa jadhr al-mal al-asli.

Wa-idha aradta an ta'khudhah thalathan, fa-Draba thalathata fi thalatha thumma fi al-mal; fa-jadhr al-natij huwa thalathatu a'da' jadhr al-mal al-asli.

Wa-idha aradta an tajida niSf jadhr al-mal, fama 'alayka illa an naDriba niSf fi niSf, wa-huwa rub'; thumma hadha fi al-mal: fa-jadhr al-natij sa-yakunu niSfa jadhr al-mal al-awwal.

al-Tawdih al-Riyadi:

$$2\sqrt{S} = \sqrt{4S}$$

$$3\sqrt{S} = \sqrt{9S}$$

$$\frac{1}{2}\sqrt{S} = \sqrt{\frac{1}{4}S}$$

al-tafsir: al-qawa'id hiY $\sqrt{a^2 \cdot S} = a\sqrt{S}$. al-Khwarizmi yuqaddimuha ka-Khwarizmiyya 'amaliyya: li-tuDa'ifa al-jadhr, aDrib al-mal (mal) fi 4 (2x2) thumma khudh al-jadhr. hadhihi al-tariqa tatajannabu al-ta'amul ma'a "naSf al-jadhr" mubashsharatan.

Chapter 6

fi al-Qisma

al-Nass al-Asli li-l-Khwarizmi

Idha aradta an taqsima jadhr tis'a 'ala jadhr arba'a, fa-bda' bi- qismati tis'a 'ala arba'a, fa-yu'tika ithnayn wa-rub'an; fa-jadhr hadha huwa al-'adad alladhi taHtaju ilayhi – wa-huwa wahid wa-niSf.

Wa-idha aradta an taqsima jadhr arba'a 'ala jadhr tis'a, fa-aqsim arba'a 'ala tis'a; fa-huwa arba'u atsa' al-wahda; wa-jadhr hadha ithnan maqsuman 'ala thalatha; ay thulthay al-wahda. Wa-idha aradta an taDruba jadhr tis'a fi jadhr arba'a, fa-Drab tis'a fi arba'a; fa-yu'tika sitta wa-thalathin; fa-khudh jadhr, fa-huwa sitta; wa-hadha huwa jadhr tis'a maDruban fi jadhr arba'a.

Wa-idha aradta an taDruba Di'fa jadhr tis'a fi thalathati a'da' jadhr arba'a, fa-khudh Di'fa jadhr tis'a Hasaba al-qawida al- madhkura, Hatta ta'rifa jadhr ayi mal huwa. Waf'il al-mithla bi-l-nisbati li-l-thalathati juthur al-arba'a li-ta'riffa ma yajib an yakuna mal dhalika al-jadhr. Thumma aDrab hadhayni al-malayn ahadahuma fi al-akhar, fa-jadhr al-natij yusawi Di'fa jadhr tis'a maDruban fi thalathati a'da' jadhr arba'a.

Qawa'id al-Juthur 'inda al-Khwarizmi:

$$\frac{\sqrt{9}}{\sqrt{4}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1.5$$

$$\frac{\sqrt{4}}{\sqrt{9}} = \sqrt{\frac{4}{9}} = \frac{2}{3}$$

$$\sqrt{9} \times \sqrt{4} = \sqrt{9 \times 4} = \sqrt{36} = 6$$

$$(2\sqrt{9}) \times (3\sqrt{4}) = \sqrt{36 \times 36} = 36$$

al-qawa'id al-'Amma:

- $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$
- $\sqrt{a} \times \sqrt{b} = \sqrt{a \times b}$
- $(m\sqrt{a}) \times (n\sqrt{b}) = mn\sqrt{ab}$

Hadhihi al-qawa'id hiya al-asas alladhi yabni 'alayhi al-Khwarizmi jami' 'amaliyyatihi 'ala al-juthur.

Chapter 7

al-sittu Masa'il al-TawDiHiyya

al-Nass al-Asli li-l-Khwarizmi

Qabla fuSuli al-Hisab wa-anwa'iha al-mukhtalifa, sa'uqaddimu al-an sitta masa'il, kamithal 'ala al-HaSalat al-sitt allati tanaWalat ha fi bidayat hadha al-'amal. Qad bayyantu anna thalathan min hadhihi al-HaSalat, li-Halliha, la taHtaju ila naSf al-juthur, wa-dhakartu aydanna anna al-Hisab bi-l-jabr wa-l-muqabala yajib ann yaqulaka da'iman ila ihdha haa al-HaSalat. Sa-ulHaqu hadhihi al-masa'il al-an, allati sa-takunu aqraba ila al-fahm, wa-tusahhilu istiya' al-mawdii', wa-tubayyinu al-ijj b-iDaaa akbar.

limadha sittu masa'il?

Yuqaddimu al-Khwarizmi ba'da shariH al-HaSalat al-sitt **sitta masa'il tatbiqiyya** – mas'ala wahida li-kulli HaSala. Hadhihi al-masa'il laysat majarrad tamariyn riyadiyya, bal hiya *qiSas* (word problems) ta'ksi mawaqif waqi'iyya. Wa-al-ahammu min dhalika annaha tuZhiru kayfiyyat **nimuDajat** mas'ala waqi'iyya ila mu'adala jabriyya – wa-hiya al-mahara al-asasiyya fi 'ilm al-jabr.

al-mas'ala al-ula

al-Nass al-Asli li-l-Khwarizmi

“Qasamtu 'ashara ila qismayn; Darabtu ahadahuma fi al-akhar; ba'da hadha Darabtu ahada al-qismayn fi nafsih, wa-hasil al-Darb fi nafsih arba'atu a'da'i ma hasala min Darbi ahadi al-qismayni fi al-akhar.”

al-Hall: li-yakun ahad al-qismayn shay'an, wa-al-akhar 'ashara naqis shay'. TaDrabu al-shay' fi 'ashara naqis shay'; fa-huwa 'asharat ashya' naqis mal. Thumma taDribuhu fi arba'a, li-anna al-mas'ala taqulu “arba'atu a'da'.” Fa-l-natij arba'un 'asharat ashya' naqis arba'ati amwalin. Ba'da hadha taDribu al-shay' fi al-shay', ay ahada al-qismayni fi nafsih. Wa-hadha mal, huwa musawin li-arba'in shay'an naqis arba'ati amwalin. KhaffiDha al-an bi-l-arba'ati amwalin, wa-azidha ila al-mali al-waHid. Fa-l-mu'adala: arbau 'ishruna shay'an tusawi khamsata amwla; wa-malu wahid yusawi thamaniyata juthuran, ay arba'a wa-sittin; wa-jadhr hadha thamaniya.

al-Hall al-Riyadi:

$$x^2 = 4x(10 - x) = 40x - 4x^2$$

$$\Rightarrow 5x^2 = 40x \Rightarrow x^2 = 8x = 64 \Rightarrow x = 8$$

al-qismaan huma: $x = 8$ wa $10 - x = 2$.

Hadhihi al-mas'ala taquluna ila ihdha al-HaSalat al-sitt, ay: **mal yusawi usuLan.**

al-mas'ala al-thaniya

al-mas'ala: "Qasamtum 'ashara ila qismayn; Darabtu kulla qism fi nafsihi, wa-ba'dahu 'ashara fi nafsaha; hasilu 'asharat fi nafsiha yusawi ahada al-qismayni maDruban fi nafsihi, wa-ba'dahu fi ithnayn wasab'at atsa'."

$$100 = x^2 \times 2\frac{7}{9} = x^2 \times \frac{25}{9} \Rightarrow x^2 = \frac{900}{25} = 36 \Rightarrow x = 6$$

al-qismaan huma: $x = 6$ wa $10 - x = 4$.

al-mas'ala al-thalitha

al-mas'ala: "Qasamtum 'ashara ila qismayn. Qasamtum al-wahida 'ala al-ukhar, wa-al-natij arba'a."

$$\frac{10 - x}{x} = 4 \Rightarrow 10 - x = 4x \Rightarrow 10 = 5x \Rightarrow x = 2$$

al-qismaan huma: $x = 2$ wa $10 - x = 8$.

al-mas'ala al-rabi'a

al-mas'ala: "Darabtu thulthi shay' wa-dirham fi rub'i shay' wa-dirham, wa-al-natij 'ishrun."

$$\left(\frac{x}{3} + 1\right) \left(\frac{x}{4} + 1\right) = 20 \Rightarrow \frac{x^2}{12} + \frac{7x}{12} + 1 = 20 \Rightarrow x^2 + 7x = 228$$

naSf al-juthur: 3.5. al-murabba': 12.25. nuDil ila :228 240.25. al-jadhr: 15.5. nanquS 3.5: $x = 12$.

al-mas'ala al-khamisa

al-mas'ala: "Qasamtum 'ashara ila qismayn; ba'da Darbi kulli qism fi nafsihi, jamatuhuma; wa-al-majmu' thamaniya wa-khamsun dirhama."

$$x^2 + (10 - x)^2 = 58 \Rightarrow 2x^2 - 20x + 100 = 58 \Rightarrow x^2 + 28 = 11x$$

$$x = \frac{11}{2} \pm \sqrt{\frac{121}{4} - 28} = 5.5 \pm 2.5. \text{ al-qismaan: } x = 3 \text{ aw } x = 7.$$

al-mas'ala al-sadisa

al-mas'ala: “Darabtu thultha jadhr fi rub'a jadhr, wa-al- natij yusawi al-jadhr wa-arba'a wa-'ishrina dirhama.”

$$\frac{x}{3} \cdot \frac{x}{4} = x + 24 \Rightarrow \frac{x^2}{12} = x + 24 \Rightarrow x^2 = 12x + 288$$

$$x = 6 + \sqrt{36 + 288} = 6 + 18 = 24$$

Chapter 8

masa'il mukhtalifa

al-Nass al-Asli li-l-Khwarizmi

Wa-hadhihi masa'il iDafiyya li-tawsi' al-fahm wa-ta'kidi al-mabadi' allati buniyat 'alayha al-HaSalat al-sitt al-kanuniyya.

masa'il tawDiHiyya iDafiyya

al-Nass al-Asli li-l-Khwarizmi

mas'ala :1 qasamtu 'ashara ila qismayn, wa-Darabtu ahadahuma fi al-akhar, fa-kana al-natij wahidan wa-'ishrina.

mas'ala :2 qasamtu 'ashara ila qismayn; ba'da Darbi kulla qism fi nafsihi, naqastu al-asghar min al-akbar, wa-al-baqiyy arba'un.

mas'ala :3 lana murabba', thulthahu min khumushihi yusawi sub'i jadhr.

mas'ala :4 wujud jadhr murabba', idha Duriba fi arba'a amthalihi kana al-majmui' 'ishrina.

mas'ala :5 murabba', idha Duribat arba' min juthur fi khamsat min juthur antajat Di'f al-mal, ma'a fa'iD sitta wa- thalathin dirhaman.

mas'ala :6 wujud murabba', idha uDifa thulthuhu ila thalathati darahim, wa-nuqiSa al-majmu' min al-murabba', thumma Duriba al-baqiyy fi nafsihi, astarada al-murabba'.

Hullu-l-masa'il al-mukhtalifa:

mas'ala :1 $x(10 - x) = 21 \Rightarrow 10x - x^2 = 21 \Rightarrow x^2 + 21 = 10x$
 $x = 5 \pm \sqrt{25 - 21} = 5 \pm 2 = 3 \text{ aw } 7$

mas'ala :2 $(10 - x)^2 - x^2 = 40 \Rightarrow 100 - 20x = 40 \Rightarrow x = 3$

mas'ala :3 $\frac{2}{3} \cdot \frac{1}{5}x^2 = \frac{1}{7}x \Rightarrow \frac{2}{15}x^2 = \frac{1}{7}x \Rightarrow x = \frac{15}{14} = 1\frac{1}{14}$

mas'ala :4 $x \cdot 4x = 20 \Rightarrow 4x^2 = 20 \Rightarrow x^2 = 5 \Rightarrow x = \sqrt{5}$

mas'ala :5 $4x \cdot 5x = 2x^2 + 36 \Rightarrow 20x^2 = 2x^2 + 36 \Rightarrow 18x^2 = 36 \Rightarrow x^2 = 2$

mas'ala :6 $(x - \frac{x}{3} - 3)^2 = x \Rightarrow (\frac{2x}{3} - 3)^2 = x$
 $\Rightarrow \frac{4x^2}{9} + 9 - 4x = x \Rightarrow x^2 + 20\frac{1}{4} = 11\frac{1}{4}x \Rightarrow x = 9 \text{ aw } 2\frac{1}{4}$

Part I

al-Tatbiqat al-'amaliyya

al-jabr wa-al-Hayat al-yawmiyya:

Min ahamma ma yumayyizu kitab al-Khwarizmi anna al-jabra 'indahu laysa mujarrad lu'bat ithniyya aw niZariyya mujarrada, bal huwa **adatun 'amaliyya** li-l-ta'amul ma'a al-masa'il al-yawmiyya. YatanaWal al-juz' al-thani min kitabihi tatbiqat 'amaliyya fi thalathat majalat ra'iyya:

- .1 **al-mu'amalat al-tijariyya** – al-bay' wa-l-shira' wa-l-Sarf wa-l-ijara
- .2 **al-misaHa wa-l-handasa** – misaHat al-ardiyn wa-Hufri al-qanawat
- .3 **al-mirath wa-l-wasaya** – ahkam al-fara'id al-islamiyya

Wa-hadha ya'ksi al-wiDifa **al-'amaliyya** li-l-'ilmi fi al-HaDara al-islamiyya – lam yakun al-'ilmu munfasi-lan 'an al-Hayat, bal kana fi khidmati al-insan mubashsharatan.

Chapter 9

fi al-mu'amalat al-tijariyya

al-Nass al-Asli li-l-Khwarizmi

fi al-mu'amalat al-tijariyya: i'lam anna jami'a mu'amalat al-nas min bay' wa-shira' wa-muSara'fa wa-ijara, fa-innaha la takhlu min qadrayni wa-arba'ati a'dadin yas'alu 'anha al-sa'il, wa-hiya al-miqdar wa-l-thaman wa-l-kayfiyya wa-l-jami'. 'adad al-miqdar yunasibu 'adad al-jami', wa-'adad al-thaman yunasibu 'adad al-kayfiyya. thalathatu min hadhihi al-arba'a da'iman ma'luma, wa-Wahid majhul, wa-huwa alladhi yatadammanuhu qawluhu "kam?" wa-huwa al-maTlub.

Qa'idat al-arba'a a'dad – al-jabr al-tijari:

Yasifu al-Khwarizmi huna ma nusammi al-yawma "qawa'id al-thalath" (Rule of Three) aw "al-tanasiq". Kullu mas'alat tijariyya tatadammanu arba'a kamiyyat murtabiTa bi-l-tanasiq:

- **al-miqdar** (al-miqdar): kamiyyat al-sila' fi al-Safqa al-ula
- **al-thaman** (al-thaman): si'r tilka al-kamiyya
- **al-kayfiyya** (al-kamiyya): kamiyyat al-sila' fi al-Safqa al-thaniya
- **al-jami'** (al-majmui'): al-si'r al-ijmali al-maTlub

thalathu min hadhihi al-kamiyyat ma'lumma wa-l-rabi'a majhula. Wa-l'alaqa hiya: al-miqdar : al-kayfiyya :: al-jami' : al-thaman aw bi-l-mu'adala: $a \times b = c \times d$

Qa'idat al-arba'a a'dad:

al-miqdar	al-thaman
a (ma'lum)	b (ma'lum)
c (ma'lum aw majhul)	d (ma'lum aw majhul)

$$\text{al-majhul} = \frac{\text{hasil Darb al-'adadayni al-mutanasibayn}}{\text{al-'adad al-ma'lum al-thalith}}$$

al-Amthila:

.1 "ashara bi-sitta, kam bi-arba'a?" al-kamiyya = $\frac{10 \times 4}{6} = \frac{40}{6} = 6\frac{2}{3}$

.2 "ashara bi-thamaniya, ma yakunu al-jami' li-arba'a?" al-jami' = $\frac{8 \times 4}{10} = \frac{32}{10} = 3\frac{1}{5}$

.3 "lil-'amili ujrati 'asharati darahim fi al-shahr; kam yajib an takunu ujrathu fi sittati ayyaam?"
al-ujra = $\frac{10 \times 6}{30} = 2$ dirahm, ba-frD anna al-shahr thalathun yawman.

Chapter 10

al-misaHa wa-l-handasa

al-Nass al-Asli li-l-Khwarizmi

i'lam anna ma'na qawlina “wahid fi wahid” huwa al-misaHa: ay Dhiraa' fi Tuliha fi Dhiraa' fi 'arDiha.
kullu murabba' mutasawi al-aDla' wa-l-zawaya, lahu Dhiraa' wahida li-kulli Dil', fa-lahu wahid fi misaHihi. Fa-in kana li-mithali hadha al-murabba' Di'lan fi Dil'ih, fa-inna misaHat al-murabba' 'arba'atu amthali misaHa murabba' Dil'uhu Dhiraa' wahida.

manhaj al-Khwarizmi fi al-misaHa:

Yabda'u al-Khwarizmi bi-ta'rifi al-misaHa 'ala annaha murabba' al-wahda (Dhiraa' fi Dhiraa'). hadha al-ta'rif yaj'alu al-misaHa **kamiyya mujiba da'iman**, wa-huwa ma yufassiru iqtisaru 'ala al-ma'amilat al-mujiba fi al-jabr.

al-mulahadha al-muhimma anna al-Khwarizmi yafhamu al-misaHa *bi-shakl hindisi* – fa-l-misaHa laysat majarrad 'adad, bal hiya murabba' lahu Tul wa-'arD. hadha al-tafakkur al-hindisi huwa alladhi yumakkin-uhu min i'tai al-burahin al-hindisiyya li-l-HaSalat al-jabriyya.

10.1 al-muthallathat

al-Nass al-Asli li-l-Khwarizmi

Idha Darabta irtifa'a ayi muthallath mutasawi al-aDla' fi niSfi al-qaa alladhi yaqumu 'alayha al-khaTT al-mamayyiz lil-irtifaa' 'amudiyyan, fa-l-natij yu'ti misaHa dhalika al-muthallath.

misaHat al-muthallath:

$$\text{misaHat al-muthallath} = \text{al-irtifaa'} \times \frac{\text{al-qa}}{2}$$

hadhihi al-Sigha SaHiHa li-kulli muthallath, laysa faqaT al-muthallath al-mutasawi al-aDla'. wa-l-Khwarizmi yastakhdimuha fi masa'ili misaHa al-ardiyn.

10.2 al-dawa'ir

al-Nass al-Asli li-l-Khwarizmi

Fi kulli da'ira, hasilu Darbi qiTriha fi thalath wa-sabi'a wahida yusawi al-muHit.
 Wa-tujad misaHa ayi da'ira bi-Darbi niSfi al-muHit fi niSfi al-qiT.
 Wa-idha Darabta qiTa ayi da'ira fi nafsiki, wa-naqasta min al-natij sub'an wa-niSfa sub'in minhu, fa-l-baqiy yusawi misaHat al-da'ira.

misaHa wa-muHit al-da'ira 'inda al-Khwarizmi:

$$C = \pi d \approx \frac{22}{7} \times d \quad (\text{al-muHit})$$

$$A = \frac{C}{2} \times \frac{d}{2} = \frac{Cd}{4} \quad (\text{al-misaHa – al-tariqa al-ula})$$

$$A = d^2 - \frac{d^2}{7} - \frac{d^2}{14} = d^2 \times \left(1 - \frac{3}{14}\right) = d^2 \times \frac{11}{14} \quad (\text{al-misaHa – al-tariqa al-thaniya})$$

mulahadha: qimmat $\pi \approx \frac{22}{7}$ hiya al-taqrib al-shahir alladhi istakhdamahu al-riyadiyyun al-islamiyyun wa-l- Yunaniyyun. Amma al-tariqa al-thaniya fa-yuHassibu fiha al-Khwarizmi $\pi/4 \approx 11/14$, wa-huwa taqrib mumtaz.

10.3 al-aHjam

al-Nass al-Asli li-l-Khwarizmi

Hajam al-jism ruba'i al-aDla' yujad bi-Darbi al-Tul fi al-'arD, thumma fi al-irtifaa'.
 Wa-amma al-makhruTat wa-al-ahram, kayna al-muthallath aw al-ruba'i, fa-tuHassabu bi-Darbi thuluth misaHa al-qa fi al-irtifaa'.

Hisab al-aHjam:

$$V_{\text{muwazi al-mustatiilat}} = \text{al-Tul} \times \text{al-'arD} \times \text{al-irtifaa'}$$

$$V_{\text{haram}} = \frac{1}{3} \times (\text{misaHa al-qa}) \times \text{al-irtifaa'}$$

hadhihi al-Sighar SaHiHa wa-la tazalu tudarRasu fi al-madaris Hatta al-yawm!

10.4 al-muthallath al-qaaim al-zawiya (mubarhanat fifaghorus)

al-Nass al-Asli li-l-Khwarizmi

laHiz annahu fi kulli muthallath qaaim al-zawiya al-Dil'ayn al- qSirayn kulla wahid minhuma maDruban fi nafsih, wa-majmu' al- hasilayn, yusawi hasila Darbi al-Dil' al-Tawil maDruban fi nafsih.

mubarhanat fifaghorus 'inda al-Khwarizmi:

$$a^2 + b^2 = c^2$$

Haythu a wa b huma al-Dil'aan al-qSiran (al-saqan), wa c huwa al-watar (al-Dil' al-Tawil).

al-burhan al-hindisi: yuqaddimu al-Khwarizmi burhanan hindisiyyan jamilan bi-istikdam murabba' maqsum ila arba'a murabba'at Sagira mutasawiya wa-thamani muthallathat mutasawiya, mubayyinann anna majmu' murabba'ayi al-Dil'ayni al-qSirayn yusawi murabba' al-Dil' al-Tawil.

Chapter 11

fi al-wasaya wa-'ilm al-fara'id

al-Nass al-Asli li-l-Khwarizmi

YatanaWalu al-juz' al-akhir min hadha al-'amal ahkam al-mirath al-islami (al-fara'id). Wa-hadha al-mawDu', ma'a annahu yabdu min jinsi al-fiqh la al-riyadiyyat, yaHtaju li-jami' adawat al-jabr fi Hallihi Hina turakkabu al-wasaya 'ala al-as-ham al-shar'iyya.

al-jabr wa-l-fara'id:

YuZhiru hadha al-faSl – alladhi ya'tabiruhu ba'Du al-mu'arrikhin akthara li-l-ih-timam fi al-kitab – kayfa anna al-jabr kana **adatan Darruriyya** Hatta fi al-'ulum al-shar'iyya. fi al-qanun al-islami li-l-mirath, hunaka *as-ham thabita* muHaddada fi al-qur'an al-karim (mithla: li-l-zawja al-thumnu, li-l-abi al-sudus, ila akhirihi). Lakin 'indama yuDafu waSiyya (taTaww'iyya) ila hadhihi al-as-ham al-thabita, taSbiH al-mas'ala *jabriyya*: naHtaju li-majhul x (li-yumaththila al-mablagh al-ijmali aw al-waSiyya. Hadha huwa bi-ZabT ma daafa' al-Khwarizmi li-ta'lifi kitabihi – fa-l-quDat wa-l-fuqaha' kanu bi-Haja li-Tariqa munazZama li-Halli hadhihi al-masa'il.

fi ras al-mal wa-l-duyun

al-Nass al-Asli li-l-Khwarizmi

mas'ala: "tawaffa rajulun wa-taraka ibnayn, wa-awSa bi-thulthi ras mali-hi li-ajjanibi. Taraka 'asharati darahim malan wa-'asharati darahim dainan 'ala ahadi al-ibnayn."

al-Hall: tusammiya al-mablagh alladhi yustakhraj min al-dayni shay'an. tuzilha ila ras al-mal alladhi huwa 'asharatu darahim. a-yakunu al-majmuu' $10 + x$. TanquSu minhu thulthan, li-annahu awSa bi-thulthi mali-hi: $3.33 + 0.33x$. al-Baqiy $6.67 + 0.67x$. Taqsuma hadha bayna al-ibnayn: yakulu kulla wahid minhuma $3.33 + 0.33x$. Wa-hadha yusawi al-shay' al-maTlub.

al-Hall al-Riyadi:

$$3.33 + 0.33x = x \Rightarrow 3.33 = 0.67x \Rightarrow x = 5$$

YaHulu al-ajjanibi 'ala 5 darahim; wa-yaHulu kulla walad 'ala 5 darahim (yabqii al-ibnu al-madynun bi-daynihi kamilan ka-muqasSa Hissatuhu).

al-mu'adala: nahnu nabHathu 'an x alladhi yumaththilu al-mablagh alladhi yuKHsamu min al-dayni li-yuzafa ila al-tarika qabla qismat al-waSiyya.

fi naw' akhar min al-wasaya

al-Nass al-Asli li-l-Khwarizmi

mas'ala: "tawaffa rajulun tarikan ummahu wa-zawjatahu wa-akhawayn wa-ukhtayn lahu min ab wa-umm, wa-awSa li-ajnanibi bi-tsa'i ras mali-hi."

al-Hall: tuj'alu as-hamu al-tarika min thamaniya wa-arba'in juz'an. Wa-ta'lamu annaka idha akhadhta ts'an min ayyi ras mali yabqa thamaniya atsa'ih. azid al-an ila al-thamaniya atsa' thumnaha minha, wa-ila al-thamaniya wa-arba'ayn aydan thumnahun, ay sitta, li-ikmal ras mali-ka. Fa-hadha yu'ti arba'a wa-khamsin. yaHulu al-shakhS alladhi yu'Sa lahu bi-ts'i 'ala sittatin min hadha. al-baqiyya thamaniya wa-arba'un tuqSamu bayna al-warathah Hasaban li-as-hamihim al-shar'iyya.

al-Hall al-Riyadi:

al-warith	al-sahm (min 48)
al-umm	8
al-zawja	6
kullu akh	14
kullu ukht	7
al-majmu'	$8 + 6 + 2(14) + 2(7) = 56$ juz' mu'addal

al-Khwarizmiyya: ras al-mal = .54 al-waSiyya = 6 (ts' al- .54 al-baqiy = 48 yuqSamu bayna al-warathah. Hadhihi al-mas'ala tataTallabu wijda maDami' mushtarak (48) yumkinu an yastaWi 'ala jami' al-as-ham al-shar'iyya ba'da izalat al-waSiyya.

masa'il al-wasaya al-murakkaba

al-Nass al-Asli li-l-Khwarizmi

mas'ala: "tawaffat imra'atun tarikatan zawjaha wa-bnan laha wa-ummaha. awSatat li-shakhsin bi-thuluthay maliha, wa-li- akhar bi-rub'ih. j'alat al-waSiyyatayn ma'an 'ala ibniha, wa-'ala ummiha niSf (Hissati al-umm min al-baqiyya); 'ala zawjiha la tulzimuhu illa al-thuluth (alladhi yajib an yusam bihi, Hasaba al- qanun)."

al-Hall al-Riyadi:

al-warith	al-sahm al-asli	al-taSaruf
al-ibn	$7/12$	mukallaT bi- $\frac{2}{5} + \frac{1}{4} = \frac{13}{20}$
al-zawj	$3/12 = 1/4$	yusamim bi-thulthi sahmihi = $1/12$
al-umm	$2/12 = 1/6$	mukallafat bi-niSfi sahmihi = $1/12$

al-mu'adala: al-musahimat al-ijmaliyya = $\frac{13}{20} \times \frac{7}{12} + \frac{1}{12} + \frac{1}{12} =$ muHassabat tanasubiyyan
Hadhihi al-mas'ala al-murakkaba tuZhiru al-quwwa al-Haqiqiyya li-l- jabr fi Hall al-masa'il al-shar'iyya
al-murakkaba allati kana yuwajihuha al-quDat.

Appendix A

mulakHHas al-HaSalat al-sitt

al-HaSalat al-sitt al-kanuniyya li-l-mu'adlat al-tarb'iyya:

al-HaSala	al-ism al-Arabi	al-mu'adala	al-Hall
al-ula	mal yusawi usuLan	$ax^2 = bx$	$x = b/a$
al-thaniya	mal yusawi 'adadan	$ax^2 = c$	$x = \sqrt{c/a}$
al-thalitha	usuL tusawi 'adadan	$bx = c$	$x = c/b$
al-rabi'a	mal wa-usuL yusawi 'adadan	$ax^2 + bx = c$	$x = \sqrt{(b/2a)^2 + c/a} - b/2a$
al-khamisa	mal wa-'adad yusawi usuLan	$ax^2 + c = bx$	$x = \frac{b}{2a} \pm \sqrt{(b/2a)^2 - c/a}$
al-sadisa	usuL wa-'adad yusawi malan	$bx + c = ax^2$	$x = \frac{b}{2a} + \sqrt{(b/2a)^2 + c/a}$

mulahadha tarikhiyya: Hadhihi al-HaSalat al-sitt tastanfudhu jami'a imkaniyyat al-mu'adlat al-tarb'iyya dhat **al-ma'amilat al-mujiba**. al-iqtiSir 'ala al-ma'amilat al-mujiba kana Darruriyyan bi-sabab al-tafsir al-hindisi: fa-l-Khwarizmi yafhamu kulla Haddin fi al-mu'adala 'ala annahu miqdar hindisi (masaHat wa-Tul), la yumkinu an takuna al-misaHa saliba. wa-l-mu'alaja al-kamila li-jami' al-mu'adlat al-tarb'iyya bima fiha al-ma'amilat al-saliba lam ta'ti illa fi 'asr al-nahDa al-urubbiyya.

Appendix B

mu'jam al-muSTalaHat al-jabriyya al-Arabiyya

al-muSTalaH al-Arabi	al-nuskha al-latiniyya	al-ma'na wa-l-tafsir
al-jabr (al-jabr)	algebra	idafat ma yanquSu min ahadi Tarfay al-mu'adala ila al-Taraf al-akhar li-istikmal al-tawazun
al-muqabala (al-muqabala)	almuchabala	hadf ma yatashabahu min al-Tarfayn li-tashil al-mu'adala
al-jadhr (al-jadhr)	radix	al-majhul dhatuhu x (– “al-shay' ” al-majhul
al-mal (al-mal)	census	mirab' al-majhul x^2 (– “al-tharwa” aw “al-mumtalakat”
al-'adad al-mufrad (al-'adad al-mufrad)	numerus simplex	al-Hadd al-thabit fi al-mu'adala (al-'adad al-ma'lum)
al-shay' (al-shay')	res/cosa	al-majhul x (– kalimat badlla li-l-jadhr
al-dirham (al-dirham)	drachma	wahidat al-nuqd/ al-qiyas – tus-takhdam kawwahida li-l-a'dad
al-waSiyya (al-waSiyya)	legatum	al-tarika aw al-waSiyya fi qanun al-mirath
al-fara'iD (al-fara'iD)	legata	al-as-ham al-ilzamiyya fi al-mirath al-islami
misaHa (misaHa)	mensura	misaHa al-ard – Hisab al-masaaHat
Darb (Darb)	multiplicatio	'amaliyyat al-Darb
qisma (qisma)	divisio	'amaliyyat al-qisma
jam' (jam')	additio	'amaliyyat al-jam'
TarH (TarH)	subtractio	'amaliyyat al-TarH

al-muSTalaH al-Arabi	al-nuskha al-latiniyya	al-ma'na wa-l-tafsir
jadhr (jadhr)	radix	al-jadhr al-tarb'i
muka"ab (muka"ab)	cubus	muka"ab al-majhul x^3

tatawwur al-muSTalaHat al-jabriyya:

'inda tarjamat al-latiniyyin li-kutub al-riyadiyyat al-arabiyya fi al-qarn al-thani 'ashar al-miladi (fi Tulay-Tula wa-Siqilliyya), Hawwalu al-muSTalaHat al-arabiyya ila latiniyya:

- **al-jabr** → **algebra** (min "al-jabr")
- **al-shay'** → **res** (al-shay'/al-jawhar) → **cosa** (fi al-iTaliyya) → **coss** (fi al-almaniyya)
- **al-mal** → **census** (al-mumtalakat)
- **al-jadhr** → **radix** (al-jadhr)
- **al-dirham** → **drachma** (al-'umla)

Wa-kalimat "**jabr**" (algebra) hiya al-waHda allati waSalat ila al-lugha al-inkliZiyya mubashsharatan min al-arabiyya – wa-hiya iHda al-kalimat al-qallla dhati al-aSl al-arabi fi al-riyadiyyat al-inkliZiyya ila janib "**Sifr**" (zero, min al-Sifr) wa-"**jayb**" (sine, min al-jayb) wa-"**Khwarizmiyya**" (algorithm, min al-Khwarizmi).

mulahadha blbliyughrafiyya

al-maSadir al-makhTUTa:

al-makhTUTa al-ra'isiyya li-kitab al-Khwarizmi *al-mukhTasar fi Hisab al-Jabr wa-l-Muqabala* maHfUza fi maktabat Bodleyan bi-Oxford (makhTUTa Hunt. ,214 awraq ,(123–1 nusikhat sanat 743 H / 1342 M. Wa-hiya juz' min majlud CMXXVIII alladhi yaHtawi ayDan 'ala thalath muSannafat ukhra fi al-Hisab wa-l-jabr.

al-makhTUTa maktuba bi-KhaT wadiH wa-maqrū', lakinnaha taniqu ila Haddin kabiirin min al-nuqaT al-i'jamiyya allati tumayyizu al-Huruf al-mutashabiha.

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